KSTVET Kenya School of TVET

The Home of Technical Education

PHYSICAL AND HEALTH EDUCATION DEPARTMENT

BALLGAMES SKILLS 1 AND ATHLETICS PE B 7101

By the end of the course the learner should be able to:

i. Explain the aims and objectives of Physical education and the importance of physical activities.
ii. Narrate in detail the historical development of Football, Basketball, volleyball, Handball and Athletics
iii. Derive enjoyment and relaxation from participating in Football, Basketball, Volley ball, Handball and Athletics.
iv. Acquire and develop skills and techniques and apply them for participation in Football, Basketball, Volleyball, Handball and Athletics
v. Participate in the interclass Football, Basketball and Volley ball, Handball and Athletics competitions.
vi. Acquire skills, attitudes and knowledge needed by a sports officer/PE teacher
vii. Participate as an official in a Football, Basketball, Volleyball, Handball and Athletics during inter class competitions.
viii. Apply the knowledge learnt to promote sports in the society.
ix. Apply first aid principles to treat injuries during sports.

INTRODUCTION

Qualities of a good P.E. TEACHER

Should possess the following components of Physical fitness:

- Endurance
- Speed - Co-ordination
- Flexibility - Power
- Strength - Reaction time
- Agility - Balance

Should be knowledgeable with the subject
Should be cheerful
Should be confident
Should be sympathetic
Should be understanding
Should have good communication skills
Should be approachable
Should be organized

Duties/responsibilities of a P.E teacher

Instructor - Team Manager/Chaperon
Counsellor - Demonstrator
Dietician - Administrator
Coach - First aider
Trainer - Referee/umpire
Evaluator – Disciplinarian

Problems encountered when teaching P.E

1. Lack of role models
2. Lack of resource materials e.g. books
3. Lack of proper facilities e.g. fields, gymnasiums
4. Lack of equipment e.g. balls
5. Lack of proper game uniforms
6. Competition from examinable subjects
7. Negative attitude by pupils, fellow teachers and parents
8. Conflict with religion

9. Conflict with tradition and cultural beliefs

Purpose of P.E. to the nation

 Enables people to perform skilful and efficient movement through use of physical and mental power.

 People use movement as a means of communication, expression etc

 People use leisure time well in creating, performing and appreciating physical activities e.g. games, dance e.t.c

 Promotion, preservation and appreciation of national and international culture.

 Self-discipline through rules and regulation

 Learning and participating in P.E creates awareness of natural resources and environment.

 People learn to improvise and use equipment in various activities.

 Maintenance of high standard of body health, strength, fitness and general body development.

ATHLETICS

INTRODUCTION

Running has flourished as sport form time immoral, even before the first recorded Olympic Games in 776BC. The other athletic disciplines (events), however have developed over many centuries through man's natural instinct to compete against others in throwing, throwing, or jumping as high as possible as jumping as far as possible.

Modern track and field is amalgam of many different activities. Those events which did not begin in Greek times have in most part growing form the contests in rural medieval Europe, and reflect their parenthood in their rules. Wall leaping in Cumberland and wheel throwing in Ireland gave rise to "Polevault and "Hammer throw". Discus throwing and javelin throwing with their metric specification reflect their continent.

During the tale nineteen century local interpretations of the rules were modified, firstly in universities where competitive athletic became organized, and later through the development of the international and Olympic movements, and by 1912 had assumed their present-day form. Few changes have been made since.

The sport however is still developing, with the throwing events beginning to outgrow the confines of the arena, and more events becoming accepted as suitable for women. In recent times we have witnessed the introduction of 400m hurdles for women, 1500, and 3000m races also. Women even ran in marathon.

It is important that teachers continue to install to the young ones in school the spirit of sports for the good of their health and pride of their nation.

TACK EVENTS

These are events done on the track. They are-:

(i) Short races i.e. sprints

(ii) Middle races

(iii) Long races

SPRINTS/ SHORT RACES

They are

- 100m
- 200m
- 400m

These races are run in lanes.

To succeed in these races the competitors must build ability to move their limbs more quickly powerfully and officially than others. This can only, be achieved through hours of diligent practice

(a) race of leg striking
(b) Force and rang of the rear leg drive- the extended rear leg kicks hard for a forward drive. The upper leg (thigh) makes almost a 90 to the torso. The strides are wide but controlled
(c) Balance between arm and leg movement"- the head is over leading. The arms are just pulled to a point infront the chest and not infront of the face.
(d) Relaxation and freeline. The whole body books relaxed – this resphely the general mechanism of the body e.g. circulation, respiration and maculation.

2. THE START OF SPRINTS

(a) Competitors draw for lanes
(b) Starting blocks may be used but they must conform to the stipulated specifications e.g.

(i) They must be adjustable but must be without sprints of other devices to give artificial aid to the runner
(ii) They must be constructed entirely of rigid materials
(iii) They must be so constructed that both feet are in contact with the ground while starting.
(iv) Starting block must be approved by track referee.

Figure 1. An image of a starting block

(c) While starting the competitor observes the following points

(i)The position of the body

"On your marks" position, the competitor takes correct and comfortable position. Hands are outstretched and spread just behind the starting line. The head is leading, the strong leg should be placed in the front block since during stating, and this is the leg which does most work.

THE START

For years, the three standard sprints have been.

i. **Elongated start**- where in the "take your marks" the knee of the rear leg is level with or slightly behind the heel of the foot front.

ii. **Medium start**- the knee of the rear leg is placed opposite a point in the front half of the front foot.

Figure 2. A demonstration of a medium start

iii. **Bunch start** – where the toes of the rear foot are abuts twelve inches to the toes of the lead foot.

Sprinters generally develop starting positions which fall into these general categories and develop efficient starts by experimentation during practices. The actual feet spread is determined be

successful starting. The front foot is usually placed from twelve to eighteen inches from the starting line. In general, taller men (usually using and elongated start) place the front foot abut eighteen inches away from the line. Short men using a bunch or a modified bunch start move the front foot closer to the starting line.

SPRINTING

In the "get set" position the arms are straight and the weight is equally distributed between the front foot and the hand. The hips should be raised at least to the horizontal with very close foot placement (bunch start) a higher hip even inch necessary.

Best results are gained by keeping the back leg relaxed and bringing it through fast and low, concentrating on the powerful drive the lead let.

It is a mistake for an athlete in the "get set" position to look the down the track by raising his head no purpose is served and it leads to lenience in the neck and shoulders, and also make the athlete come up too quickly out of his blocks. The face should be practically parallel with the track and the eyes should be fixed five or six yards down the track.

Figure 3. A set position in sprints

In the "get set" position the sprinter must concentrate on the move and not the gun.
Constant practices with the gun helps develop an automatic reaction to the sound.

Competent starters will hold the athlete in the "get set" position for the required two seconds. The athlete must be absolutely motionless at that line.

THE RACE

The runner must concentrate his attention on the exposition off the marks. He accompanies the powerful drive form the lead leg with vigorous arm action, as the body drivers forward.

The trailing (rear) must come forward rapidly, with the foot just barely clearing the ground. The ease speed and power with which this leg takes the first determines initial acceleration.
The athlete must **not drive** up this line. He should reach at erect running position too soon.
The left are compensates for the rapid pull-through of the right leg.
The first sleep from the blocks, usually onto the right foot, must be natural for the athlete. A very long first step checks the forward momentum. A very short slop causes a loss of balance.

SPRINTING

The arm action above all else should be natural and relaxed. In ideal form the forward swing brings the sprinter's hand almost to shoulder height. The rear swing brings the hand back of the hip. The elbow is approximately at a right angle and fixed but not rigid. The arms should swing at the sides in line with the run and should not go across the chest so far that the shoulder and touch would be turned.
The toes must point directly forward at all line. The necessary bounce or rebound found in great splinter would course be impossible in the athlete who toes out rather that forward.

On the "Set" position, the camp should raise the hips to just above shoulder. This brings shoulders slightly forward and not completely over the hands. The neck is relaxed. The eyes are focused at about 3 meters down the track.

"Gun fires" This provides a release. The gun is fired then the competitors are completely motionless. The riving leg and ankle must be fully extended and force must be directed more horizontally than vertically. The drive form blocks should be strong and arms should move.
A competitor maintains horizontal dive for with to twelve strides gradually coming upright with each stride.

When running in lance the competitor keeps his lane throughout the competition.
Changing of lanes entails disqualification.

3. THE FINISH

The “dip finish” (pushing the chest toward the large) is good if only correctly done.

The most recommended finish is where a competitor maintains a running form right through the tape.

The order in which the competitor’s torso reach the nearer edge of the finishing line determines the finishing position.

Figure 4. ‘Dip finish’ illustration

MIDDLE DISTANCES

They include 800m, 1,500m and 3,000m (steeplechase) races
A middle distance runner must be predominantly interested in improving the following:-

LONG DISTANCE RACES

5,000m race and 10,000m, consider points as for middle races, but the training is related to the actual event

HUDRLING

100m, 110m, 400m

A hurdle should consist of two upright supporting a rectangular frame and should have a level top rail. The hurdle may be adjustable in height but must be rigidly flattened at the required height for each event. The hurdle shall conform to the following specification

Overall width 120cmc
Length of bas 70cms
Depth of top bar 70cm
Thickness of top bar 10-25m
Minimum total weight 10kgs

The toppling force required to overturn a hurdle should to between 3kg and 4kg for adults, 2.7kgs and 3kgs hurdle for juniors under 17 years. The top bar should be striped black and while.

Figure 5. A hurdle

(b) When hurdling

(i) The hurdler should have exact number of strides between flights in most cases 8 strides in high hurdles and 15-18 in low hurdles

(iii) He should be relaxed in his movement but should be concentrative in his action especially toward the next barrier.

(iv) Flight over the hurdle is a continuous section with least time lost

(v) The hurdler keeps his body low over the hurdles and keeps a forward body lean throughout the landing

(vi) Where international hurdles are used competitor may knock over any number of berries (hurdles) during the race without penalty. where these type of hurdles are not used a competitor may knock a maximum of three without disqualification

(vii) A hurdle is not allowed to trail his foot or leg around the side of the barrier while clearing it.

HURDLE SPECIFICATIONS

Race distance	**Age group**	**Height of barrier**	**No. of barriers**	**From start to the first barrier**	**Interval between barrier**	**Distance last meddle to finish**
MEN'S						
110m	Adults	10.70cm	10	13.72m	9.14m	14.02m
110m	<19 years	99.00cm	10	13.72m	9.14m	14.02
100m	<17 years	91.40cm	10	13m	8.5m	10.5m
400m	Adults	91.4cm	10	45m	35m	40m
WOMEN'S						
100m	Adults	83.80cm	10	13.0m	8.5m	10.5m
80m	<17 years	96.20cm	8	12.8m	8m	12.0m
400m	Adults	76.2cm	10	45m	35m	40m

7 STEEPLE CHASE

3000m

The steeplechaser must combine the strength and stamina of the 500m runner with speed of 1500m, runner, and the skills of a hurdler.

The hurdles are made of heavy timber so that they are not overturned easily. They should be **91.4cm**, high with tolerance of 3mm above and below. Each should be 3m wide and have a top bar of 12.7cm.

At the water jump the hurdle must be firmly fixed. The water obstacle shall be 3.66m sqare in plan and have as elevation which slopes form **70cm** deep at the hurdle and to the ground level at the further.

Figure 6. A steeplechase barrier

(a) WHEN JUMPING

(i) Competitors are allowed to jump, vault or place one foot (stepping) on the barrier while negotiating it.

(ii) A keen c competitor steps on the barrier with on foot and on to the shallow water with the other foot.

(iii) The competitor concentrates as the approaches the barrier

(iv) The keeps his body low as he riots over this supporting let

(v) He places the sole of the supporting foot firmly on top of the hurdle

RELAY RACES

4 X 100m

4 X 400m

(a) BATON EXCHANGE

(i) This is cone within the box i.e. 20m box in 4X100m, 10m for 4X400m.

(ii) Visual L-R –L-R i.e. Receiving the baton with left-hand transferring and carrying it with right-hand.

(iii) non visual carrying the baton with the same receiving hand i.e the baton is never transferred form on hand to the other in the process of running

(iv) In all cases the competitors does not see the baton, but feels it when receiving. He receivers it while in speed.

(v) The receiving runner should give enough space within the lane for his partner to run on during change over process

Figure 5. Baton exchange in relays There are several variations of baton passing in the sprint relays but the only basic differences are to be found in the position of into receiver " hand and in the incoming runners method of handing off(placing) the baton.

The responsibility of flitting the bar get (the outgoing runners extended hand) is palaces on the incoming man, as the relief man is placing forward when the pass is made.

In this pass the outgoing runner must hold this palm up, for the baton, is placed in his hand a downward motions.

PART 11

FIELD EVENTS

To all field events especially throws, precautionary measures must be exercise to minimize chances of injuries.

1. HIGH JUMP

There are four different style of high jumping namely: scissors, western roll, straddle and fosbury flop.

The last two are the most recommended to beginner. The best performance has been achieved on these styles, although western roll has been dominantly popular for many years.

(a) Straddle technique

(i) Approach (run) Approach the bar in a straight line at an angle of about 35.

- A maximum of a steps distance is sufficient for the man
- Have definite distance

(ii) Take off - be accurate with the make off foot the point of takeoff is specific

- The last 3 strides become more and more flat footed
- Plan the take off foot about 1 meter for the up right. There is a slight backward learn to gain force from the trunk and hips for the lift. Keep the trunk upright.

(iii) The flight - try to drop the leafing leg and should shown when you are at bar level

(iv) Landing: landing is on the back or on shoulder and upper back

(v)

(b) Fosbury flop (flop)

Figure 7. fosbury flop technique

(i) Approach (run) Approach the bar form a curved run. Improve the curved run to a walking stick run

- The curved run takes you to 1 meter from the uptight
- The force of the curved run turns your back toward the bar as you bread contact with the ground. The curve should be maintained to give rotation movement over the bar

(ii) Flight: - Permit your leg to hand loose immediately you take off

- Tuck the keels toward the head. The head looks backward toward the landing area.

(c) Western roll technique

(i) Approach: - Approach the bar in a straight line at an angle of about 40. - Be accurate on check point i.e. on take-off mark

(ii) Take off - Land the take-off foot on the take-off mark knee and hip slightly bent.

(iii) The flight - Lands stretched

- The take-off leg is stretched as it clears the bar

(iv) The landing - 3 point landing i.e. the take-off foot and the hand

(d) Scissors technique

This is a most uneconomical style of jumping. This is because of the amount of body weight to be lifted above bar level in order to clear the lowest part of the body

Figure 8. Scissors technique demonstration

2. RULES OF HIGH JUMP

Height in high-jump heights is measured perpendicularly from the ground to the lowered point of the cross- bar A competitor fails if he:-

1. Dislodge the bar from the peg
2. Touches the landing area beyond the place of the upright a without first boarding the bar
3. Takes off from both feet

<u>Ties are decided as follows: (TIE BREAK)</u>

1. The competitor with the lowest number of jumps at the higher he place. **<u>Height at which the tie occurs</u>** shall be awarded the higher place.
2. If the tie still remain, the competitor with he lowest number of jumps (whether successful or not) throughout the competition up to and including the height last cleared shall be awarded the higher place
3. If the remains the competitor with the lowest total number of failures throughout the competition up to and including the height last cleared shall be awarded a higher place
4. If the tie remains

(i) If for **<u>first place</u>**, the bar shall be raised or lowered by 1 ½ in or half the range it was raised, until the winner is decided

(ii) If for any other place the competitors shall be given the **<u>same place</u>**

Central rules

1. The qualifying height is decided by officials before the competition.
2. The competitor may opt to jump or not to jump when called for his turn.
3. The bar is never lowered for a competitor who misses a jump.
4. A competitor is allowed 3 attempts in any one height
5. A competitor is discontinued form further jumping when he makes a consecutive failures at one height
6. A competitor who fails to clear any height, even the qualifying height, is not **awarded any place**.

The table below show soring and tie breaking in high jump

	5,6”	5,8”	5,9”	5,10	5,11”	6,0”	6,1”	TOTAL FAILURES	TJ OUTMAPLS	POSITION
	1,67	1,72	1,75	1,77	1,80	1,82	1,85			
OPIYO	-	xo	Xo	xo	o	Xxo	xxx	8	6	2
KAMAU	-	xxo	Xo	xo	xo	Xo	xxx	9	6	3
LETING	O	O	Xo	Xo	xxo	Xxo	xxx	10	6	4
MUTISO	-	-	O	xO	xxO	Xo	xxx	7	6	1
NGETHE	Xxx	-	-	-	-	-	-	-	-	-

O = Cleared height, X failed jump and – unattempt

D. COMPETITION RULES

The rules of high jump shall apply in addition to the following rules:-

1. The competitor falls if he touches the ground with his pole beyond the plane of the uprights.
2. If he leaves the ground for the purpose of making a vault and fails to clear the bar.
3. He fails if, he makes 3 runs without completing a vault
4. He fails if, in the process of vaulting be climber the pole If the pole breaks during vaulting the competitor shall be given another chance.

3 LONG JUMPS

(a) Essential of long jump

(i) Fast approach which maintains speed right into the take off

(ii) Powerful vertical lift

(iii) A flight technique which permits efficient landing position

(b) APPROACH

(i) Have a definite approach run distance. International jumper will require about 21 to 28 strides to take off. Long jumpers will need 15-19 strides.

(ii) Have a fast run but controlled on

(iii) Accelerate onto the take off-board

<u>Take off</u>

(i) Hit the board with the correct foot, stike up through your hips punching your free knee and opposite arm, skywards

(ii) The chest and hips should push forward to ensure maximum lift

(c) LIGHT AND LANDING

(i) Fast hind leg swaung to lead off the board, the arms and behind the trunk

(ii) Cycling action in the air

(iii) When landing, feet are brought together and stretched forward heels to touch the ground first.

(iv) Head is kept low and forward. Hands sweep forward to avoid sitting backward

(d) COMPETITORS

(i) Each competitor is given 3-6 jumps he is credited on his best jump

(ii) Incase of a tie the 2nd best jump decides the winner if the tie cannot be diced and it concerns

(iii) 1st place, the competitors tying shall be given extra jump

(iv) If the tie concerns other positions the competitors shall be given same position

(v) At competitor shall take off form the stepboard and not beyond

(vi) A competitor must land within the landing area only

(vii) Measurements are taken at right angles front be scratch line to the nearest break in the sand made by many part of the competitor. The measurements are taken to the nearest ¼ below the distance covered

4 TRIPLE JUMP

(a) Essentials in triple jumping

(i) Gain maximum distance during each phase of jump without prejudicing the following phase

(ii) Place each foot landing so that it given efficient position form which to jump

(iii) The flight of top is kept low. The ratio of 12:10:12 is best recommended

(b) Approach

(i) Have a definite approach run distance

(ii) Have fast but controlled open

(iii) Have up and accelerate on to the takeoff board.

(c) TAKE OFF

(i) Take off form the step board

(ii) kept the trunk and head upright and drive hips forward and downward away from the board

(d) Flight and landing

(i) Increase the step distance by swinging the free leg high

(ii) In the step keep the heal of the lead-knee well in “wait the ground to come to meet you, rather than seek it with your front leg”

(iii) When landing buist sideways to avoid losing distance by setting back into the sand

(d) Competition rules

All rules of long jump apply:-

5. DISCUS

(a) The grip (The hold) Discus is usually held and not griped

(i) Hold the discus loosely in the hand so that the end joints of your fingers wrap around the discus in

(ii) Stand/ stance

(iii) Stand at the back of the circle facing away from the direction of the throw

(c) The swing

(i) When the wind up is completed transfer your weight over to the turning foot (ii) Keep turning on your toes until the foot bearing your weight is pointing in the direction of the throw.

(ii) The turning from the back of the circle to the front part should be smooth and momentum

(iii) Keep the turn close to the ground throughout

(iv) Do no jump

(d) RELEASE

(i) The throwing arm in extended behind the shoulder

(ii) the throwing hip is head of the throwing arm at the centre of the circle

(iii) As the weight is transferred from the near foot to font toot, the front foot turns towards the direction of throw.

(iv) Both feet must remain incontact with the circle until the discus has left the hand

(v) The throwing arm is kept fully extended throughout the throw

(e) Follow through

After the implement leaver hand the thrower keeps on turning on the throwing leg to a stop

(f) Competition rules

(i) The discus is thrown from the circle to any point between the two sector lines

(ii) The competitor should not tough any part of the line of the circle in the process of throwing

(iii) After throwing the competitor shall leave the circle form the rear half and in a standing position only

(iv) A foul throw and letting the implement for in an attempt has landed

(v) The competitor shall leave the circle only when the implement has landed.

(vi) The measurements shall be taken from the adage of the mark made by the implement with is nearest to the scratch line through the centre of the circle

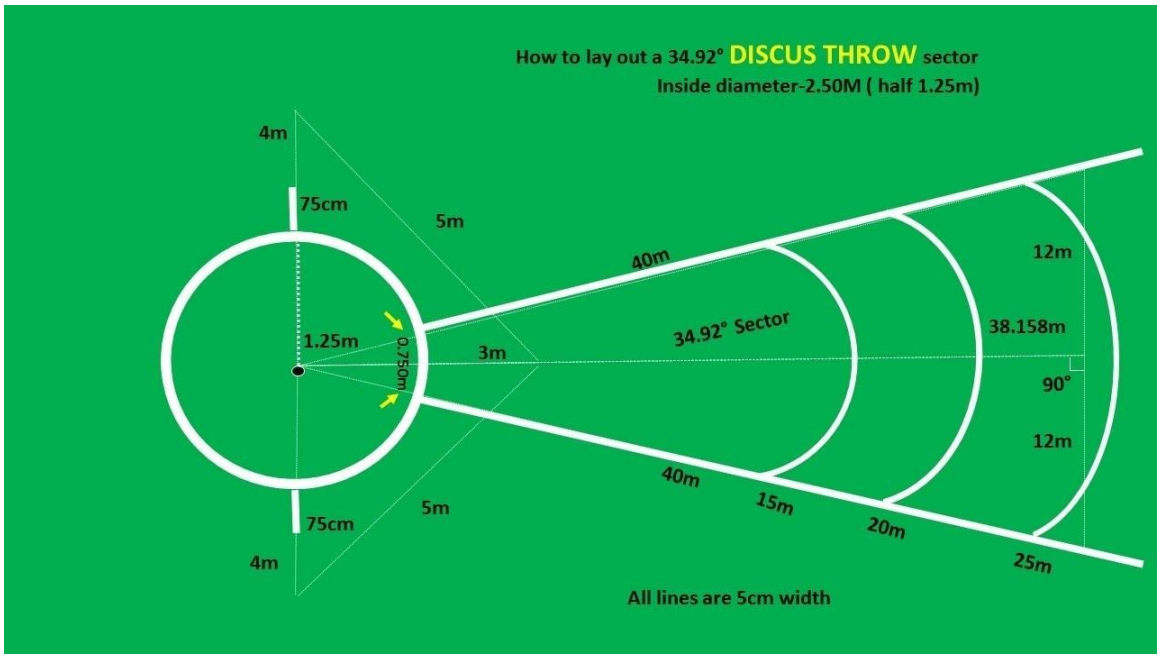


Figure 9. A discus playing field

Weight for discus

Men: weigh (2kg) Diameter (219mm)

Women: Weight (1kg)

BoysWeight 1.5kg – 1.75 kgm

6 SHOT-PUT

(a) Stance

(i) In rotational method the throw face back of the throwing area. In conventional the thrower faces the throwing area

(ii) Be relaxed

(iii) Feet are hip width apart, and left arm high

(b) The hold (grip)

(i) Hold the shot against the middle three fingers with be thumb and little finger giving lateral support. (The short is actually proposed and not gripped)

(ii) The short must touch or be inclose proximity to neck and must remain there until release

(iii) Right elbow is well out

(c) Shift (glide) and release

(i) Start at the back of the circle facing away from the direction of throw, and with on right leg.

(d) Follow though

(i) Keep the shot moving forward

(ii) Drive the right hip up hard and chase the shot out with your finger tips.

(iii) Drive the right hip up ward and chase the short out with your finger tip

(iv) Keep balance using reverse – foot action effected by transfer of whole wight from back foot to front foot.

(e) Competition rules

(i) The competitor shall be allowed 3-6 throws and shall be credited on this best throw

(ii) Incase of a tie, the 2nd best throw shall decide the winner. If the tie remains the next best shall decide the winner. When the throw dies not decide the positioning the competitor are placed in one position unless the tie concerns 1st position of

(iii) The weight shall be put from the shoulder with one hand only and not behind one place of shoulder

(iv) the weight must fall within the edges of the sector lines

(v) The competitor is not allowed to touch any part of the circles in the process of throwing but however, he may touch the inside part of the stop board.

(vi) The competitor must not leave the circles until the implement has touched the ground and then he must leave only from the rear half of the circle, in a standing position

(vii) The measurement is taken from the edges of the mark which is nearest to the scratch line. The distance is near off at the inside of encircle. Measurement is taken to the nearest 1 ½ inch fellow the distance covered.

(viii) The measurement is taken from the edge of the mark: which is nearest to the secretion line. The distance is near off at the inside of hecircle. measurement is taken to the nearest 1 ½ inch below the distance covered

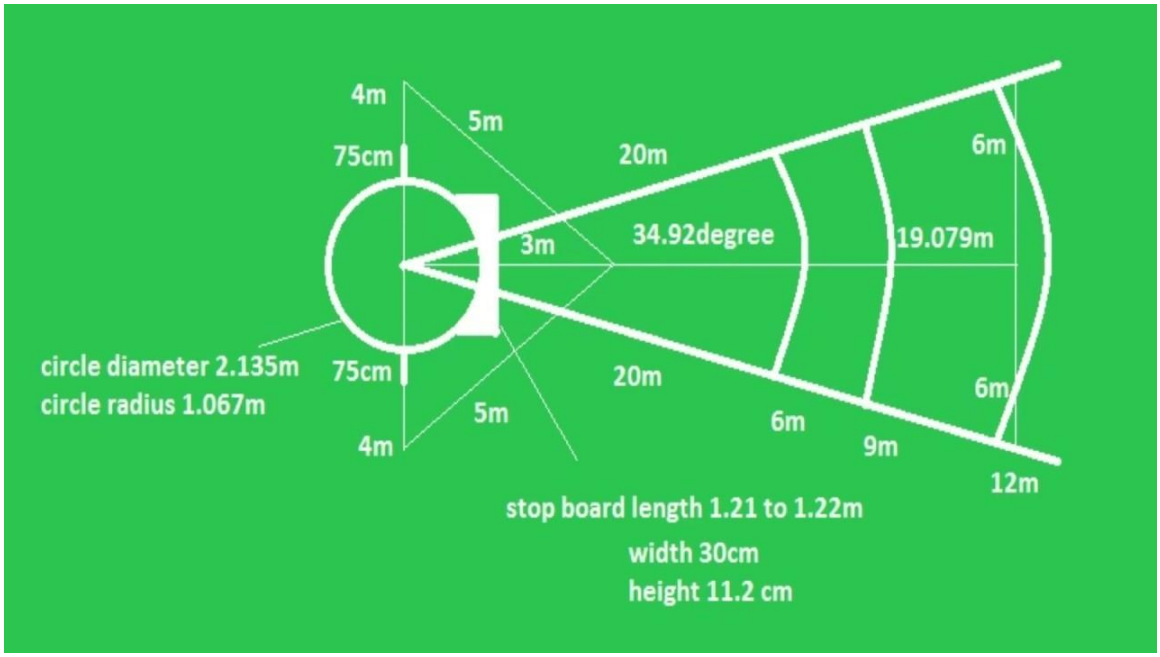


Figure 10. A shot-put playing field

7. JAVELIN

(a) The grip

(i) The javelin rests along the palm of the hand

(ii) The finger or fingers and thumb, through which force is applied, pull against the near edge of the binding

(iii) The other fingers should spread out along the grip and they should hold firmly

(B) THE RUN UP/ APPROACH

(i) During the run the javelin is carrying above the shoulder with the palm facing upwards. The elbow is kept under the javelin

(ii) The javelin is carried pointing slightly upwards or even horizontally. The grip is kept firm and javelin in position throughout

(C) THE RELEASE

(i) Javelin is withdrawn back into its throwing position behind the throwing shoulder

(ii) The throwing leg takes a long stride landing on the heal inform of the body

(iii) The whole weight is transferred form the near foot to the front foot

(iv) The hand remains high. The javelin is pulled form a point slightly behind the shoulder. The elbow passes close to the ear, and the hand outsetched for a release. The body turns side – ways to avid crossing of the front line.

(D) THE LANDING

(i) The angle of projection is approximated 40° - 50°.

(ii) The javelin lands point/the head first

(iii) The javelin must land in the sector of 28.96°

(E) PRECAUTIONARY MEASURES

In order to avoid accidents competitors must be given instruction Hal that, implements must be thrown during proper practice or competition only. After throwing, the implement must be returned by hand not thrown back to the stating area.

The referee or official concerned shall disqualify form competing in the vents, and athlete who willfully disobeys the above instructions having his a attention drawn to them

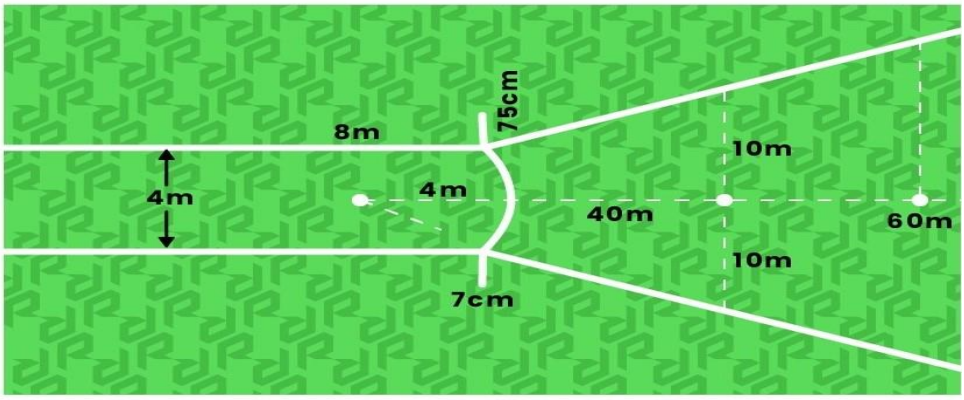


Figure 11. A javelin playing field

RULEs OF COMPETITIONS

(i) A foul throw or letting go of the implement in an attempt to throw shall be reckoned as trial

(ii) If the implement breaks in the air, it shall not be counted as a trial

(iii) The javelin shall be thrown with one hand only, and must be held at the grip.
It shall be thrown over the shoulder and it must not be along or hurled

(iv) At no time the competitor may turn round or has back towards the throwing are in the reprocess of throwing

(v) The throw shall not be measured if the javelin does not strike the ground point first

(vi) The throw shall not be measured if the competitor crosses the scratch line or its extension

(vii) The throw shall not be measured if the competitor cresses the scratch line or its extension

8. HAMMER

Hammer throwing required rotational speed, strength, coordination ability and keen sense of rhythm

(a) stance

(i) Stand behind of the circle near the edge in the rear half

(ii) Face behind the throwing area

(iii) Be relaxed. Feel should be shoulder with apart

(b) The grip

(i) Hold the handle in the palm of your left hand. You may wear gloves on your left hand to avoid formation of blisters due to friction.

(ii) The head of the hammer on its farthest position on the right side of the foot

(c) The swings

(i) Start the swing from the right side to the left

(ii) Get the longest radius by stretching arms to the full

(iii) Bend the knees but keep the trunk straight and upright

(iv) Move the hips to maintain balance during the swings(“heel-toe “ turns)

(d) The turn

Three swings are enough to start the turn

(i) Start turning as the hammer comes in front of you

(ii) The weigh from the right t toes is transferred to the left heel and the hips continue to turn

(iii) The bent knees still maintained (sigging like positions).

(e) Deliver – release

Two three turns are enough to release the mummer

(i) The release begins as a sharp lift with your legs, finished off by your arms

(ii) Stretch arms to the full face the hammer as it files

(iii) The balance is maintained as the right foot follows through for another step

(f) Rules of competition

(i) A competitor is allowed 3-6 throws and is credited on his best throw, incase of a tie, the rule of tie shall apply a competitor is not allowed to tough the circle line or the ground

outside the circle during the process of throwing. However, the head of the hammer may touch the ground when swinging

(ii) It is a foul for a competitor to stop throwing and begins again

(iii) At competitor leaves the circle after the hammer has landed. He shall leave through the rear half of the circles and from a standing position

(iv) Measure each throw to the nearest ¼ inch below the distance covered. It the wire breaks in the air the competitor shall be given another chance

The throwing circle is 2.135m in diameter. There must be protection cage

Weight

Senior men: 7.26kg

Senior women: 4kg

The total length of the chain and the hammer is 117.5 to 121.5 cm.

Figure 12. an image of hamer

HANDBALL

Team size
This is a game played by two teams where a team consists of **14** players with **7** players in the court any one time and 7 players remaining substitutes. One of the 7 players in the court is designed to be the goalkeeper.
For a player or team official to be allowed to participate:
- He must have his/her name entered in the score sheet.
- Should be present at the start of the game.

In case of lateness of a player or officials and his name entered into the score sheet.
The minimum number of players to start the game is 5 and it can go down as the game progresses. At no time shall a team pay without a GK
NB: if a GK is suspended
1 a court player can change into GK uniform and play in his position as a GK
2 utilizes the substitute GK
3 A player goes out on behalf of the suspended GK

Substitution
A substitute become a court player when the player he is substitution leaves he in turn enter the court. A substitute can enter the court as many times as possible **revolving** substitution rule governing substitution.

Duration of the game.
The game consists of 2 halves of 30 minutes each and an interval of 10 minutes

PLAYING THE GAME
Referee inspects the court, players and the equipment.
He conducts the toss up and the captain winning the toss up choose either to start the game with first throw off or side

Timing starts at the referee signal for the initial **throw off** and ends when the timekeeper's signal released. Instances for throw off

1) Start of the game
2) After every score
3) Start of 2nd half
4) Start of extra half

Breaking a tie.
In case of a tie an extra time is added.
2 halves of 5 minutes each with break of 1 minute in between.
If it still stands duration of the same is added. If continues the tie is broken in accordance to the rules of the game.

Time off
This is a temporary stoppage of the game requested either by the captain or the coach.

When awarded
(i) When a player is injured
(ii) When coach has requested teams time out.
(iii) A stoppage for penalty throws
(iv) When player is given suspension

Equipment used in the game of handball.

- Ball
- Upright posts
- Cross bar
- Net

Score sheet

The following information are entered in the score sheet

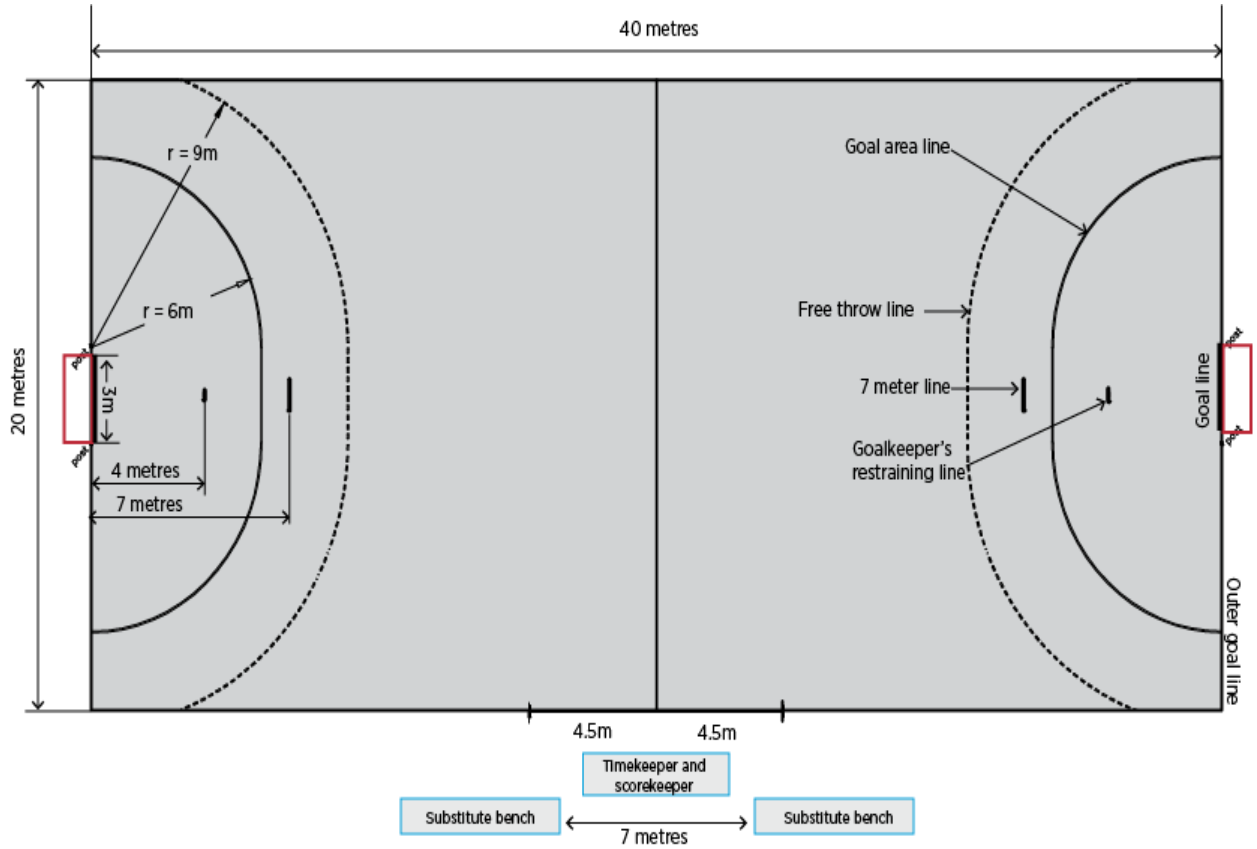
(a) Name of players and team officials.
(b) Chronological order of the scores made by each team.
(c) Punishment given to each player e.g. suspension, disqualification or exclusion
(d) Time at the start of the game
(e) Name of the teams.

Uniforms

- A pair of sports shoes
- Socks
- Shorts
- T-shirt

Goalkeeper always should wear uniforms different from other playmates and opponents. T-shirts shall have conspicuous numbers both back and front ranging from 1.
NB:Number 1, 12, 16 are reserved for the goalkeeper.
Captain wears an armlet on the arm.

Court dimensions



1. **Goal line**
   - Measures 20m
   - Forms width of the court
   - Where the goal posts are fixed[3m wide By 2m high]
   - Confirms the score
   - If a defender takes the ball out of its own goal line then a corner throw is awarded

Goal area

- Indicated by a radius of 6m Its GK area of operation
- He should not leave this area with the ball
- He should not enter the area with the ball from out Inside it.
- its where we have the 4m line

2. **4m line**

Also known as goalkeeper's line.
Goalkeeper shall not go beyond this line during penalty throw until the thrower shall have released the ball.

3. **6m line**

- Goal area line
- Marks the extent of the goal area
- Goalkeeper shall not move or cross the line with the ball under his control.
- Goalkeeper shall not collect the ball outside this line and comes back with it inside the area
- Penalty is awarded to the attacking team if a defender enters his goal area and by so doing gains an advantage over the opponent.
- Free throw is given to the defending team if at attacker enters opponent's goal area with or without the ball.
- Goalkeeper shall not collect the ball lying outside 6m line.
- During free throw by the attacking team the defenders position themselves along the 6m line.

4. 7m line

- Penalty mark
- During the penalty throw the thrower shall not step on or over this line before releasing the ball. In case of infringement by the thrower a free throw is awarded to the opponent at the 7m line.

5. 9m line

- Free throw line
- Along where all the free throws shall be taken for infringement by the defenders along the free throw zone.
- During the penalty throw all the players apart from the thrower and the goalkeeper shall be outside the 9m line.

6. Center line.

- Divides the court into 2 equal halves
- Along where throw off is taken
- All the players of the team taking the throw off shall be along this line.
- Along where substitution takes place

7. Side line.

- 40m long
- Forms length of the court
- Confirms a ball out of bound. A ball is out of bound when
  - Comes into contact with the ground outside the sideline
  - Comes into contact with an object in sideline
- Along where substitution takes place next to the centerline.
- Along where the officials table and the substitution bench are placed.

THROWS IN HAND BALL

1. Throw in

Awarded when the whole ball completely passes over the sideline.

Conditions to be observed

- Throw at the exact point where it when it went out
- At least one foot should step on the sideline
- Opponents 3m away from the ball.
- Must be released within three seconds
- A player cannot pass the ball to himself
- No blowing of the whistle
- Penalty throw Awarded when:
- Goalkeeper enters his goal area with the ball or takes it from the court to the goal area.
- A court player intentionally gives a back pass to his G.K in the goal area.

- A court player enters his goal area to gain advantage of the ball.
- When a clear chance of scoring has been spoiled.
- Dangerous play.
- Unwarranted whistle signal at the time when a team had a clear chance of scoring.

Conditions to be observed

- Must be done 3 sec. After the referee's whistle
- Shot must be directed to the goal
- The penalty thrower must not touch on or cross the penalty throw line before the ball leaves his hand.
- Other players to be beyond 9m line.
- The G.K should not move beyond the 4m line till the ball is released.

2. Free throw

Awarded incase of minor infringements of the rules within the court e.g illegal substitution double dribbling e.t.c Conditions to be observed

- Taken at the exact point where the infringement occurred
- Ball must be released within 3 sec.
- Opponents be 3m away from the ball.

3. **Goalkeepers throw.**

- Awarded when the goalkeeper gains control of the ball within the goal area
- A player throws the throws the ball over the opponent's own goal line.
- The ball is considered to be in play when it passes over the goal area line.
- No whistle for the throw.

4. Throw off

- At the start of the game
- Begin of the 2nd half
- If a whole ball passes in between the upright posts under the cross bar joining them (score)

5. **Referee throw**

Awarded when:

(i) There is a simultaneous infraction.
(ii) When the referee is not sure of the player who committed the offence.
(iii) Technical problems e.g. basting of the ball, external interference e.t.c
(iv) When lowering national flag.

Conditional to be observed

- Two opposing player stand facing each other hands alongside their body.
- Referee tosses the ball between the two opposing players.

6. Corner throw

Awarded when the whole ball completely crosses over the goal line having lastly been played by a defender. (But not in between the two upright posts).
It is taken at the edge of the court where the sideline meets the goal line

Passive play.

This when a team keeps the ball in their possession without making any recognizable attempt to attack or to shoot on the goal.
Penalty for this is free throw.

PUNISHMENTS IN HAND BALL

1) **Warning / Caution**.
Can be given **verbally** or through use of yellow card. It is given when:
- Time wasting.
- Wrong approach to an opponent.
- A persistent infringement of rules.

2) **Suspension**
This is **2 minutes compulsory break** given to a player due to:
(a) A repeated / persistent infringement concerning an approach to an opponent.
(b) When a player is disqualified, he shall be accompanied by 2 minutes suspension rule.
(c) Infringement of the substitution rule.
(d) Failure to release the ball or throw the ball away after being penalized.
(e) Rough play – directed to an opponent using unnecessary force.

3) **Disqualification**
This is where player is supposed to **leave the court** for the remaining **period part of the game.**
However one shall substitute after two minutes.
Occasions.
- At the third suspension.
  Endangering the life of an opponent e.g. pushing an opponent who is in air.
- Unsportsmanlike like language.
- Unregistered player entering the court/playing. It is indicated by **red card**

4) **Expulsion**
This is where a player is supposed to **leave court for the rest of the game and he/she shall not be** substituted. An expelled player shall be required to leave the team bench and vicinity of the game. It is indicated with 2 hands crossed above the head
Occasions
- Assaulting a fellow player or an official
- Spitting to a fellow player.

OFFICIALS IN HAND BALL

1. **Court referee**
- Shall start the game by blowing the whistle during the throw off.
- Shall raise his hand to confirm a score when a goal has been made.
- He shall penalize any infringement within the court and incase of any disapproval with the other referee, his decision shall be final.
- During a penalty throw he shall ensure that the thrower does not step on the line.

2. **Goal line referee**

- Confirm the score by blowing two sharp whistles when a goal is scored.
- Shall penalize any infringement along the 6m line.

Combined duties of the referees

- They shall run the game according to the laid down rules and regulation.
- They shall inspect the facilities, equipment and players before the start of the game
- They sign the score sheet at the end of the game.
- They shall inspect the score at the end of each half or any other time one feels like
- They shall make the final decision incase of any disapproval between the table officials.
- They shall penalize the players according to the laid down rules and regulations.
- They shall keep a record of score and shall confirm the score before approving the score in the score sheet.
- One of them shall minister the toss-up of the coin at the start of the game.
- Shall execute referee throw.
- They may terminate the match as need be.

3. **Time keeper**

- Shall take note of the time at the start of the game.
- Shall make a signal at the expiration of the game.
- He shall alert the court referee two minutes to the end of a half of the game.
- Shall time the two minutes suspension.
- Shall stop the game clock as instructed by the referee i.e. during time out.
- Shall take note of any time wasted e.g. incase of injuries or any other interruption and shall add that time at the end of the particular half.

BASIC SKILLS

1. **Passes**
   Similar to those used in net ball apart form over arm pass which is only used in hand ball.
2. **Shots**
   This includes **dive shot, set shot, and sling /under arm shot, standing shot, hook shot**.
3. **Dribbling**
   Constant bouncing of the ball on the ground by use of one hand.
4. **Dodging/faking/feinting**
   This is provoking the opponent to go in the wrong direction
5. **Blocking**
   Restricting movement of the opponent by using various body parts e.g. chest (frontal blocking) back (rear blocking) side (linear blocking)
6. **Catching**
   Single and double handed catching.

   Qualities of a good goal keeper
   - Courage
   - Flexible
   - Good reaction time
   - Very fast/swift

- Reasonably tall
- Disciplined
- Conversant with the rules and regulations of the games
- Dedicated/committed
- Strong
- Good grip of the ball
- Good co-ordination

SYSTEMS OF PLAY

Defense system

Man to man

A player is given a specific opponent to mark or guard.

Zonal guard.

Each player is given a specific area or region in the court to guard.

Combined cover

It is where a player is allocated an opponent as well as a area/region of the field to defend

Attacking tactics Simple first break

Only one player sprints forward to make a score.

Complex first break

More than one rushes forward to make attack.

Figure 8

This is an attack made by three players with the ball at any tine. A player follows his or her pass.

Posting

The use of a very tall player against short player. He is given high pass that the other players cannot touch then he scores.

Playing systems

I] 6:0	ii] 5:1	iii] 4:2
IV] 3:3	v] 3:2:1	vi] 3:2:1

FOOTBALL

FOOTBALL PLAYING RULES

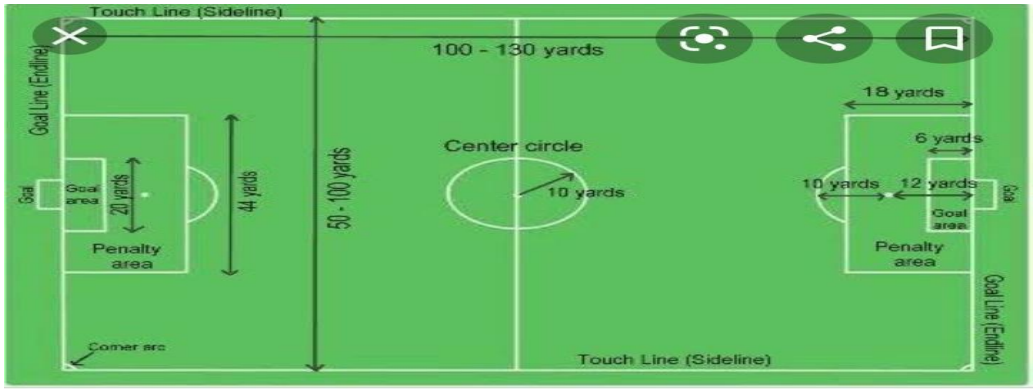
RULE 1: **number of players=11 on each side.3 substitutes allowed.**

Duration of the game-

- Normal game-2 halves of 45 minutes each .interval of 15minutes.
- Tie break-2 halves of 15 minutes each. Interval of 1 minute.

RULE 2: **the start of the game (kick off) rules:**

- Each team must have a minimum of 7 players maximum 11.
- Captains toss ends or kick off.
- The winner of the toss will choose either to start the game or choose the side on which to play from.
  Question: **why is a choice of which side to play from, an advantage in some games?**
- Players take their playing positions.
- The non- starting team must be 10 yards away from the ball i.e (outside the centre circle)
- The starting player must walt for the referee to signal the start.
- Once the ball is touched by the starting player,it is in play.
- The taker can kick for a goal and score.
- A free kick s awarded for breach of kick off rules.
  RULE NO 3: **The field of play:**



QUESTION: **What are the centre line markings of each of the markings?**

RULE NO. 4: **Goal area rules:**

- Inside here the goal keeper enjoys special protection
- G.K can catch the ball.
- Goal kick is taken from here.

RULE NO. 5: **24 yards area (penalty box).**

- Area in which the goal keeper must kick the ball for a goal keeper's kick.
- Inside here the goal keeper can catch the ball
- Inside here the goal keeper cannot receive the ball with his hands if the ball is from a back pass from his teammate: except-

a) If the ball was 'chested' to him
b) If the ball was 'headed' to him
c) If it was not an intended back pass, the ball might hit his teammate's body.
d) If a goal keeper receives a back pass with his hands, a free kick is awarded against his team(how the kick is taken will be practiced)
e) If a goal keeper crosses the box marks with the ball in his hands, a free kick is awarded against his team. It's taken from the point where the goal keeper crossed the box mark
f) Inside here the nine penal offences are penalized with a penalty. The penal offences are-
    i. A defender catching or touching the ball with his hands in the act of defence
    ii. A defender charging an opponent
    iii. A defender tripping an opponent
    iv. A defender pulling an opponent
    v. A defender obstructing an opponent.

QUESTION: **which are the other penal offences? I have given you 6, give me your 3- (an adult is not assisted to cross a river, he is shown how to cross the river)**

- If an attacker commits any of the penal offences while inside the box, a free kick is awarded against his team. The free kick is taken from where the offence occurred.
- During a penalty kick, only the taker of the penalty and the goal keeper are allowed inside the box. The rest of the players should be outside the box but inside the playing field.
- During a penalty kick the ball must be placed at the penalty mark.
  QUESTION: **how far is the penalty mark from the goal line?**

RULE 6: **the 10 yards rule.**

- It states that during a goal kick, kick off, free kick and corner kick, all opposing players must be 10 yards away from the ball.
- A spray mark is made by a referee to indicate the area beyond which opposing players should not be (the spray was introduced in the year 2016.it was called magic spray because it disappears after a minute. The kisii called it 'matiki'. We now refer to it as spray.)

QUESTION: **which are the 3 permanently marked 10 yards areas?**

RULE 7: **the goal line rule:**

- When a ball crosses this line, it is out of the field of play
- If the ball was lastly played or lastly touched by a defender (goal keeper included) a corner kick is awarded to the defender.
- If the ball crosses this line between the two goal posts and the cross bar, a goal (score) is awarded unless the playing rule have been breached.

RULE 8: **the corner kick.**

- It is awarded as explained in rule above
- The ball must be placed at the corner mark
- The opponents of the taker must be 10 yards away from the ball
- The moment the ball is touched by the taker, it is in play
- A goal can be scored directly from a corner kick
- A player can be offside directly from a corner kick

RULE 9: **the side line rule**

- When the ball crosses the side line, it is out of the field of play
- The opponent of the player who took the ball out is awarded a throw-in
- Substitution must be done along the side line

RULE 10: **the throw-in rule**

- It is awarded as explained in rule 9
- The taker of the throw must face the field
- The throw must be near the point where the ball went out through
- The ball must be thrown with both hands and from above the eyes level (the ball be above the head level)
- The thrower should not lift his feet while throwing
- The ball must be thrown and not dropped
- A score(goal) cannot be made directly from a throw-in
- A player cannot be offside directly from a throw-in

RULE 11: **the centre line and offside**

- The centre line demarcates the court into two equal halves-it shows the playing court from team A$B
- The game is started at the centre sport
- Area beyond which a player can be offside

Offside.

Offside is a situation where an attacking player places himself at an advantage position near the opponents goal line inorder to gain advantage of his position to receive the ball and score.it is not allowed in football

- A player is said to be in an offside position if: AT THE MOMENT THE BALL IS KICKED, he is nearer the opponents goal line than two defenders ie a goal keeper and more players.
- If the player has not crossed the center line, then he is not offside

Exceptions:

- If the player does not attempt to play the ball or does not gain advantage of his position, he is not penalized for offside
- If the player is merely returning to his own half, and does not attempt to play the ball, no offside
- If he receives the ball directly from a throw- in
- If he has out run his opponents and he was not originally from an offside position, he is not offside

The question to ask as a referee is where the player was at the moment the ball is kicked not after the kick.
Playing skills shall be done practically

QUESTIONS.

You now know how to cross the river, now cross by attempting the following questions:

1. **What is a dead ball situation? Give examples and proceed as a referee.**
2. **What are sudden death penalties?**
3. **Explain the substitution rule in football.**
4. **Who are the officials in a football match and what are their duties?**

- *Focus your mind on what is good (be positive)*
- *Leave corona alone, God is working on it*
- *Follow in the ministry of health guidelines*

Teams captains must toss for sides

BASKETBALL

Basketball is a team sport.

Two teams of five players each try to score by shooting a ball through a hoop elevated 10 feet above the ground.

The game is played on a rectangular floor called the court, and there is a hoop at each end.

The court is divided into two main sections by the mid-court line.

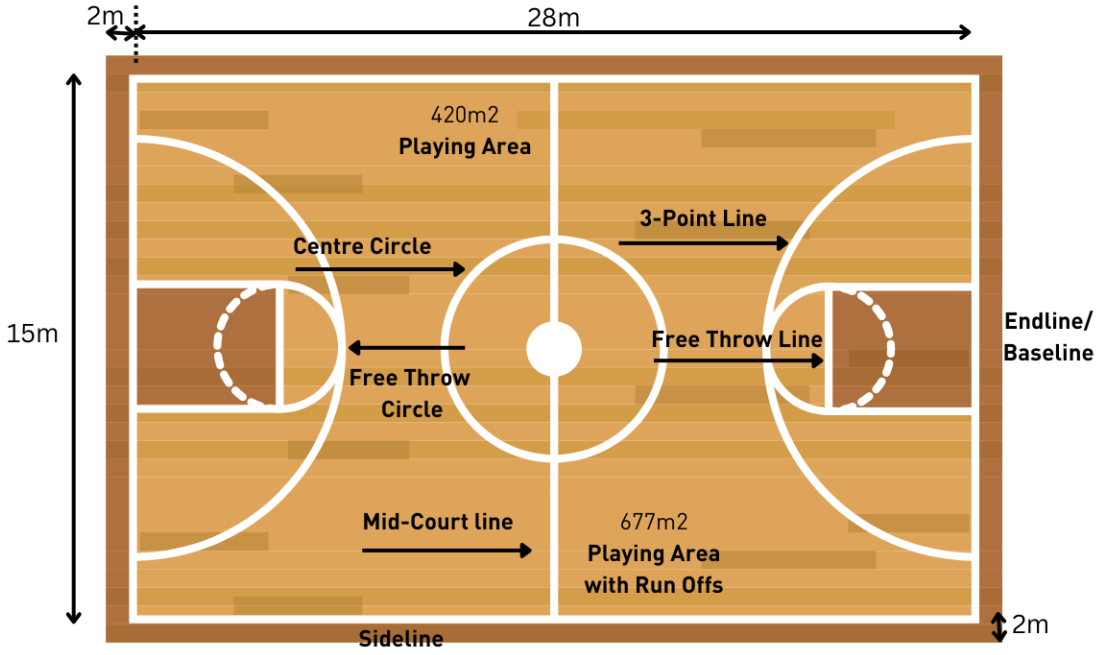
If the offensive team puts the ball into play behind the mid-court line, it has ten seconds to get the ball over the mid-court line.

If it doesn't, then the defense gets the ball.

Once the offensive team gets the ball over the mid-court line, it can no longer have possession of the ball in the area behind the midcourt line.

If it does, the defense is awarded the ball.

Basketball court



The ball is moved down the court toward the basket by passing or dribbling. The team with the ball is called the offense. The team without the ball is called the defense.

The defense tries to steal the ball, contest shots, deflect passes, and garner rebounds.

Points

When a team makes a basket, they score two points and the ball goes to the other team.

If a basket, or field goal, is made outside of the three-point arc, then that basket is worth three points. A free throw is worth one point.

Free throws are awarded to a team according to some formats involving the number of fouls committed in a half and/or the type of foul committed.

Fouling a shooter always results in two or three free throws being awarded the shooter, depending upon where he was when he shot. If he was beyond the three-point line, then he gets three shots.

Other types of fouls do not result in free throws being awarded until a certain number have accumulated during a half (called "team fouls").

Once that number is reached, then the player who was fouled is awarded a '1-and-1' opportunity. If he makes his first free throw, he gets to attempt a second. If he misses the first shot, the ball is live on the rebound.

Game Clock

Each game is divided into sections, and all levels have two halves.

In college, each half is twenty minutes long.

In high school and below, the halves are divided into eight (and sometimes, six) minute quarters. In the pros, quarters are twelve minutes long.

There is a gap of several minutes between halves. Gaps between quarters are relatively short.

If the score is tied at the end of regulation, then overtime periods of various lengths are played until a winner emerges.

Fouls and Violations

In addition to stealing the ball from an opposing player, there are other ways for a team to get the ball.

One such way is if the other team commits a foul or violation.

FOULS

Personal fouls: Personal fouls include any type of illegal physical contact.

- Hitting
- Pushing
- Slapping
- Holding
- Illegal pick/screen -- when an offensive player is moving. When an offensive player sticks out a limb and makes physical contact with a defender in an attempt to block the path of the defender.

Personal foul penalties: If a player is shooting while a being fouled, then he gets two free throws if his shot doesn't go in, but only one free throw if his shot does go in.

- Three free throws are awarded if the player is fouled while shooting for a three-point goal and they miss their shot. If a player is fouled while shooting a three-point shot and makes it anyway, he is awarded one free throw. Thus, he could score four points on the play.

- Inbounds. If fouled while not shooting, the ball is given to the team the foul was committed upon. They get the ball at the nearest side or baseline, out of bounds, and have 5 seconds to pass the ball onto the court.

- One & one. If the team committing the foul has seven or more fouls in the game, then the player who was fouled is awarded one free throw. If he makes his first shot, then he is awarded another free throw.

- Ten or more fouls. If the team committing the foul has ten or more fouls, then the fouled player receives two free throws.

Charging. An offensive foul that is committed when a player pushes or runs over a defensive player. The ball is given to the team that the foul was committed upon.

Blocking. Blocking is illegal personal contact resulting from a defender not establishing position in time to prevent an opponent's drive to the basket.

Flagrant foul. Violent contact with an opponent. This includes hitting, kicking, and punching. This type of foul results in free throws plus the offense retaining possession of the ball after the free throws.

Intentional foul. When a player makes physical contact with another player with no reasonable effort to steal the ball. It is a judgment call for the officials.

Technical foul. Technical foul. A player or a coach can commit this type of foul. It does not involve player contact or the ball but is instead about the 'manners' of the game. Foul language, obscenity, obscene gestures, and even arguing can be considered a technical foul, as can technical details regarding filling in the scorebook improperly or dunking during warm-ups

VIOLATIONS

Walking/Traveling. Taking more than 'a step and a half' without dribbling the ball is traveling. Moving your pivot foot once you've stopped dribbling is traveling.

Carrying/palming. When a player dribbles the ball with his hand too far to the side of or, sometimes, even under the ball.

Double Dribble. Dribbling the ball with both hands on the ball at the same time or picking up the dribble and then dribbling again is a double dribble.

Held ball. Occasionally, two or more opposing players will gain possession of the ball at the same time. In order to avoid a prolonged and/or violent tussle, the referee stops the action and awards the ball to one team or the other on a rotating basis.

Goaltending. If a defensive player interferes with a shot while it's on the way down toward the basket, while it's on the way up toward the basket after having touched the backboard, or while it's in the cylinder above the rim, it's goaltending and the shot counts. If committed by an offensive player, it's a violation and the ball is awarded to the opposing team for a throw-in.

Backcourt violation. Once the offense has brought the ball across the mid-court line, they cannot go back across the line during possession. If they do, the ball is awarded to the other team to pass inbounds.

Time restrictions. A player passing the ball inbounds has five seconds to pass the ball. If he does not, then the ball is awarded to the other team. Other time restrictions include the rule that a player cannot have the ball for more than five seconds when being closely guarded and, in some states and levels, shot-clock restrictions requiring a team to attempt a shot within a given time frame.

Player Positions

Centre. Centres are generally your tallest players. They generally are positioned near the basket.

Offensive -- The centre's goal is to get open for a pass and to shoot. They are also responsible for blocking defenders, known as picking or screening, to open other players up for driving to the basket for a goal. Centres are expected to get some offensive rebounds and put-backs.

Defensive -- On defense, the centre's main responsibility is to keep opponents from shooting by blocking shots and passes in the key area. They also are expected to get a lot of rebounds because they're taller.

Forward. Your next tallest players will most likely be your forwards. While a forward may be called upon to play under the hoop, they may also be required to operate in the wings and corner areas.

Offensive -- Forwards are responsible to get free for a pass, take outside shots, drive for goals, and rebound.

Defensive -- Responsibilities include preventing drives to the goal and rebounding.

Guard. These are potentially your shortest players and they should be really good at dribbling fast, seeing the court, and passing. It is their job to bring the ball down the court and set up offensive plays.

Offensive -- Dribbling, passing, and setting up offensive plays are a guard's main responsibilities. They also need to be able to drive to the basket and to shoot from the perimeter.

Defensive -- On defense, a guard is responsible for stealing passes, contesting shots, preventing drives to the hoop, and for boxing out.

Fundamentals Skills

A) Dribbling

1. Use your fingertips not your palm
2. Low stance (bend your legs) and keep the ball low
3. Keep your hand on top of the ball
4. Protect the ball with your opposite hand
5. Head up

Dribbling Violations

Traveling: Holding the ball and moving your feet in an illegal matter without bouncing the ball or pivoting. Changing your pivot foot will also result in a travel violation.

Double dribble: Dribbling the ball with both hands at the same time. It also includes picking up your dribble and dribbling again.
Carrying: When a player is dribbling and brings their hand under the ball and brings it on the top again.

B) Passing

1. Chest Pass: Use 2 hands and pass the ball from chest height to chest height to your teammate. Finish with your thumbs facing down.
2. Bounce Pass: Use the same motion as a chest pass but aim the ball about ¾ of the distance between you and your teammate at the ground.
3. Overhead Pass: Bring the ball over your head with both hands and follow through.

C) Shooting

1. Hold the ball on your fingertips
2. Have the ball resting on your shooting hand while your other hand is on the side of the ball for guidance
3. Make sure your elbow is tucked in and facing the middle of the rim
4. Bend your knees
5. Straighten your body as you go through the motion
6. Snap your elbow
7. Follow through: arm extended straight up "hand in the cookie jar"

D) Layups

Right hand layup:

1. Dribble close to the basket with your right hand.
2. Step toward the basket with your right foot.
3. Jump off your left foot
4. Shoot with your right arm as you lift your right leg
5. Aim for the sweet spot on the backboard (Left handed layups have the opposite footwork)

TIME RULES

3seconds:
Unguarded player without a ball should not remain in the opponents' restricted area for a period exceeding 3 seconds.
5 seconds:
A player should not exceed 5 seconds during throw in when the ball has been touched by the referee. A player should not hold the ball without passing, dribbling or shooting for a period exceeding
5 seconds.
A player should not exceed 5 seconds during free throw when the referee has disposed the ball.
8 seconds
A team in control of the ball should make sure that the ball moves from the backcourt to frontcourt within 8 seconds.
24 seconds

A team should make an attempt to score at the opponents' basket within a period of 24 seconds.
30 seconds
A disqualified player must be substituted within this time.
20 seconds
This is time for substitution.
1 minute
This is duration for time out.

SUBSTITUTION

A team can substitute the whole team.
Substitution is effected when court player leaves the court and the substitute enters.
Must be done within 20 seconds
A disqualified player must be substituted within 30 seconds.
Substitution cannot be done during free throw unless serious injury.

Procedure

The substitute reports to the official table and gives signal for substitution.
The substitute then sits at the substitution bench.
When the ball is dead the scorer or timekeeper signals the referee who authorizes substitution.
The substitute enters the court when the player leaves the court.

TIME OUT

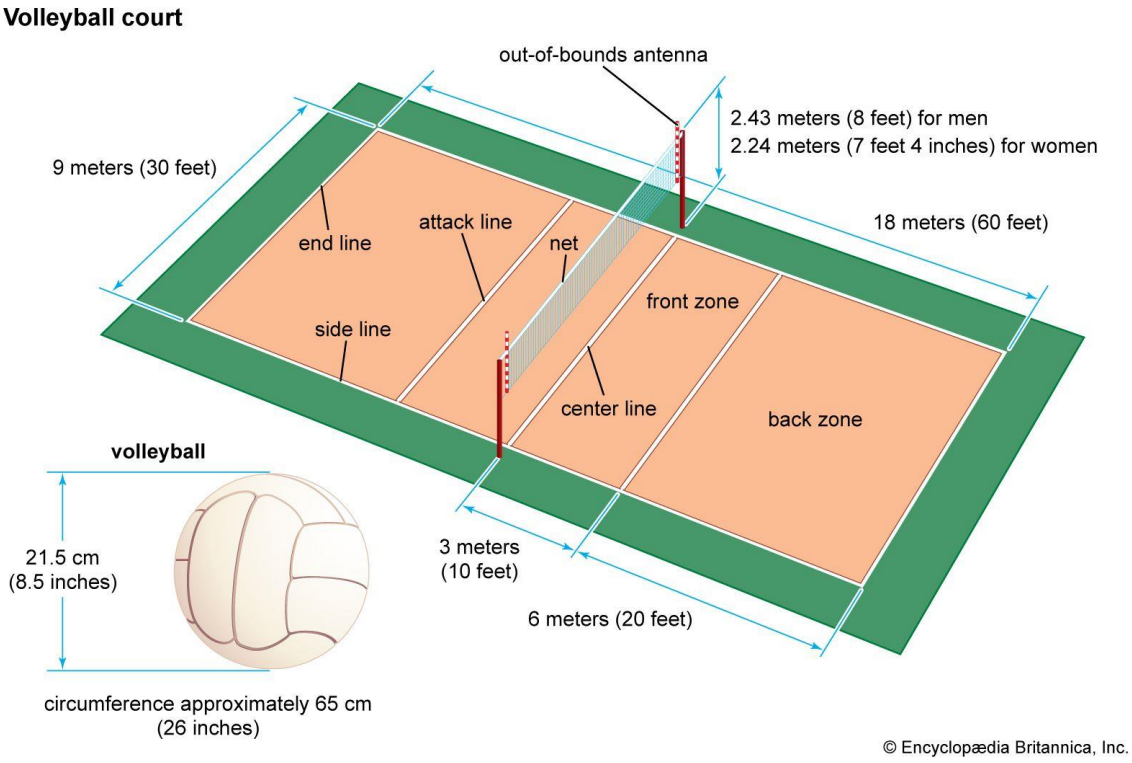
This is a period of interception of the game requested by the coach or assistant coach for technical talk.
Requested when the ball is dead.
Each team is entitled to one time out of minute between 1st, 2nd, 3rd and extra time periods.
During the 4th period 2 time outs can be given to each team.
Substitution can also take place during this time.

VOLLEYBALL

Volleyball is a team sport where two teams of six players aim to ground a ball on the opponent's court over a net.

HISTORY OF THE GAME

The game was started by William Morgan, a physical education specialist and director at Young Men Christian Association (YMCA) in 1895 in U.S.A. He first called this game Mignonette. Latter the Y.M.C.A directors suggested that the game should be called "Volley ball" because the ball was volleyed back and forth across the net.
In the early stages of development 9 players on each side were playing volley ball. The YMCA drew the first formal rules in 1897.
During the First World War the American troops introduced the game in Europe. U.S.A formed amateur volleyball association in 1928 while the Great Britain did the same in 1955. In 1946 volleyball was introduced in Olympic Games both for men and women. In 1947 the international volleyball federation was formed with its headquarters in Paris. In Kenya we have Kenya Volleyball Association (K.V.A)



LINES OF THE COURT

Side lines

Length of the court (18m)
Substitution takes place along this line
Team bench is located along this line.

End lines/Base lines

Width of the court (9m)
Service takes place along this line.

Centre line

Subdivide the court into two equal parts
Net is fixed along this line
The two referees stand along and opposite this line.
Prevents the player from entering the opponents half.

Attack line

Subdivide one of the half of the court into two parts i.e. front and back row. Front is marked 3m from the centreline.

ZONES AND AREAS

Front zone

On each court the front zone is limited by the axis of the centre line and the rear edge of the attack line. The front zone is considered to extend beyond the side lines to the end of the free zone.

Service zone

The service zone is a 9 m wide area behind each end line. It is laterally limited by two short lines, each 15 cm long, drawn 20 cm behind the end line as an extension of the side lines. Both short lines are included in the width of the service zone. In depth, the service zone extends to the end of the free zone

Substitution zone

The substitution zone is limited by the extension of both attack lines up to the scorer's table.

Libero Replacement zone

The Libero Replacement zone is part of the free zone on the side of the team benches, limited by the extension of the attack line up to the end line.

NET AND POSTS

HEIGHT OF THE NET

Placed vertically over the centre line there is a net whose top is set at the height of 2.43 m for men and 2.24 m for women.

Its height is measured from the centre of the playing court. The net height (over the two side lines) must be exactly the same and must not exceed the official height by more than 2 cm.

The net is 1 m wide (± 3 cm) and 9.50 to 10 m long (with 25 to 50 cm on each side of the side bands), made of 10 cm square black mesh.
Two white bands are fastened vertically to the net and placed directly above each side line. They are 5 cm wide and 1 m long and are considered as part of the net

ANTENNAE

An antenna is a flexible rod, 1.80 m long and 10 mm in diameter, made of fiberglass or similar material.
An antenna is fastened at the outer edge of each side band. The antennae are placed on opposite sides of the net.
The top 80 cm of each antenna extends above the net and is marked with 10 cm stripes of contrasting colour, preferably red and white.
The antennae are considered as part of the net and laterally delimit the crossing space.

POSTS

The posts supporting the net are placed at a distance of 0.50-1.00 m outside the side lines. They are 2.55 m high and preferably adjustable.
The posts are rounded and smooth, fixed to the ground without wires. There shall be no dangerous or obstructing devices.

THE BALL

The ball shall be spherical, made of a flexible leather or synthetic leather case with a bladder inside, made of rubber or a similar material.
Its colour may be a uniform light colour or a combination of colours.
Synthetic leather material and colour combinations of balls used in international official competitions should comply with FIVB standards.
Its circumference is 65-67 cm and its weight is 260-280 g

TEAM SIZE

A team is made up of 12 –layers, 6 main players and 6 substitutes.
A team may register among the top players a specialized defensive player known as libero
A team must have a captain.
The maximum number a team can substitute is 6.

LIBERO

A **libero** is a specialized defensive player in volleyball who primarily focuses on passing and digging. Here are some key points about the libero:

Position

The libero plays only in the back row and is not allowed to rotate to the front row.

Jersey

They wear a different colored jersey to distinguish themselves from other players.

Restrictions

They cannot serve, block, or attempt to block, and they cannot attack the ball if it is above the height of the net.

Role

Their main responsibilities include receiving serves and making digs to keep the ball in play.

The libero position was introduced to enhance the defensive aspect of the game and has become an essential part of modern volleyball.

START OF THE GAME

The first time referee tosses a coin between two captains and the winning captain has two options to choose:

Choice of side

The first service

The other captain takes the remaining option.

PLAYING POSITIONS

1. Right back player
2. Right front player
3. Center front player
4. Left front player
5. Left back player
6. Centre back player

SERVICE

The act of putting the ball into play by the right back player anywhere along and outside the sideline.

TYPES OF SERVICE

1. Underarm service

An underhand serve in volleyball is a technique that involves holding the ball in your non-dominant hand and striking it with the heel of your hand or your fist. It's often used by beginners because it's easier to control and return.

2. Overhead/ tennis serve

An overhead serve in volleyball is a technique where a player hits the ball over the net using the palm of their hand while their arm is raised above their head. The goal is to score points or make it difficult for the other team to return the ball.

3. A windmill is a volleyball technique where the ball is tipped or dunked over the net using finger action.

Good service.

This is when the ball is received or lands within the court.

Faulty service.

When the service is not acceptable or when the player has committed a service fault.

Examples of faulty services

Ball hitting the post

Ball passing under the net

Ball failing to reach opponents court.

Ball touching a teammate before crossing the net

Failure to toss the ball

Ball hitting the antennae

Serving the ball when standing on the end line or inside the court.

Ball landing outside the court

Exceeding eight seconds with the ball after the referee's whistle.

Playing format

To win a point

A team wins a point when the opponents make a playing fault.

Examples of playing fault

- Double hit by individual player unless there is a block.
- Four hit by a team unless there is a block.
- Positional fault (When the players are not in their playing position when the ball is being served)
- Rotational fault.
- A player touching the net with or without another opponent player
- Front court player crossing the centreline
- Ball landing inside or outside the court
- Faulty service
- Wrong substitution
- Breaking time out rules
- Back court player participating in block or spiking while in the frontcourt.
- Ball passing under the net.

Consequences of playing fault.

The serving team looses the point and chance to serve.

Receiving team gains a point and chance to serve.

The receiving team looses a point when they make a play fault thereby serving team continues to serve.

How to win a set.

In a volleyball 5 sets are played. To win a set the 1st team to reach 25 points with a leading margin of a minimum of two points takes the set. Eg 25-23, 25-21.

At the end of every set, teams changes sides. Incase of deciding (5th) set, the first team to reach 15 points with a leading margin of a minimum of two points takes the set. Team changes sides when the first team reaches 8 points.

How to win a march

The team which wins more sets emerges the winner.

BASIC PLAYING SKILLS

They include the following:

Volleying

Serving

Spiking/smashing

Blocking

ROTATION

Clockwise Movement

Players rotate one position clockwise each time their team wins a point while the opposing team is serving. This means the player in position 1 moves to position 6, the player in position 6 moves to position 5, and so on.

Substitution

This is replacement of players. It is requested when the ball is dead either by coach or the captain.

Rules governing substitution

a) The player must have registered in the line up
b) Done when the bail is dead
c) Done on the line of the first referee
d) The player to be replaced gets out first before the substitute goes
e) Requested for either by the coach or captain.
f) Maximum of 6 substitutes is allowed per set.
g) Must be conducted within the substitution zone.

Time out

- This is a brief interruption of the game or pause requested by coach or captain for a technical talk.
- Duration for time out is 30sec.
- Each team is entitled to 2 time outs per set.
- It is requested when the ball is dead.
- During time out the coach is not allowed to enter the court neither the players allowed to leave the court.

OFFICIALS OF THE GAME

1. First named referee

- He has the absolute authority over the game and other officials.
- His decision is final
- He uses whistle to signify the beginning and end of the game, a point, change of service e.t.c.
- Awarding points and services gained
- In charge of disqualification.

2. Second named referee

- Supervises control of substitution on the side of lines
- He takes note of the following:
- Net faults
- Crossing the centreline
- Illegally crossing the attack line by backcourt players.
- Reaching over the net (over reaching)
- Draws the attention of the first named referee of any unsporting behaviour.

- Notes players position at the start if the game.
- Keeps check on all time outs.
- Authorizes all substitutions.

3. The scorer

- Positions himself opposite the first named referee.
- Take note of all scores.
- He notes all time outs, substitutions and any other interruptions.
- Before the sets, he takes names of all the players, substitutes, captain, coaches.
- Take care of the positions of players in the court and ensure that the rotation order of the court is kept.
- He supplies the referee with relevant information at all times.

4. Referee assistants

- They are four and they sit at each corner of the court.
- They note the serving faults.
- Note the ball, which passes outside the net vertical markers during the service.
- Note the ball falling out of the court.

SPORTS INJURIES

NEURO INJURIES (C.N.S & BRAIN)

1. UNCONSCIOUSNESS
The state of not being able to respond to a given stimuli i.e. casualty is not aware or alert of what is happening.

Causes

- Head knock
- Shock
- Heart attack
- Poison
- Drunkard ness
- Diseases of brain
- Shortage of blood in brain
- Lack of oxygen and blood supply

Stages of consciousness to unconsciousness

i) *Full consciousness* – The casualty is not to respond to the stimuli e.g. feeling the paid when touched.
ii) *Drowsiness* – The casualty is easily aroused but relapses to unconsciousness.
iii) *Stupor* – Casualty aroused by a lot of difficulties and information given is not reliable.
iv*) Coma* – The casualty cannot be aroused at all

Treatment

Place the casualty in a comfortable position i.e. recovery/ coma position
Open the airway
Loosen the tight clothing
Resume artificial resuscitation (mouth to mouth, mouth to nose, or chest compression e.t.c)
Treat for wounds
Dispose the casualty to the hospital.

2. STROKE

Caused by blood clot blocking the blood vessels to the brain. Very common with people having high blood pressure.

Signs and symptoms

Internal bleeding shown by
Blood shot eyes
Skin becomes pale
Blood coming out through the body openings
Stool/urine contains blood stain
Blurred speech
Loss of power on sensations in arms and legs (paralyzed)

Treatment

Same as unconsciousness

3. HYSTERIA
This is caused by anxiety or too much worry

Treatment
Reassure the casualty

4. CONCLUSION
Caused by shaking up of the brain as a result of falling or direct blow.
Signs and symptoms
- Unconsciousness
- Dizziness
- Headache
- Loss of memory
- Vomiting
- Nausea
- Weak pulse

Treatment
- Place the casualty in a comfortable position
- Keep the casualty awake
- Provide a fresh air
- Give adequate rest
- Give a lot of fluids to prevent vomiting

Dispose the casualty to the hospital.
5. COMPRESSION
Caused by pressure in the brain from the fracture on the skull with the bone depressed or bleeding and swelling within the brain.

Signs and symptoms
- Confusion
- Intense headache
- Slow noisy breathing
- Paralysis on one side of the body
- Unconsciousness
- Slow strong pulse
- Unequal pupils
- Dry and flushed face
- Twitching of the limbs
- Convulsion

Treatment
- Place the casualty in a recovery position

- Stop further bleeding
- Immobilize the broken bone
- Give the casualty cool fluids, adequate rest
- Dispose the casualty to the hospital.

6. SHOCK

This is when the blood pressure is very low leading to collapsing of the body system.

Causes

- Good or bad news
- Stress
- Injuries
- Severe pain
- Nerve stimulation
- Severe illness
- Loss of the body fluids (bleeding or over sweating)
- Dehydration
- Acute abdominal emergencies e.g.
- Rapture of appendix
- Heart attack

Signs and symptoms

- Cold dumpy skin
- Sweating
- Shivering
- Confusion
- Skin becomes pale
- Fast but weak pulse rate
- Shallow breathing
- Nausea
- Vision is blurred
- Restlessness
- Coma
- Stunned attitude leading to anxiousness

Treatment

- Remove the cause of the shock
- Rest the casualty either by sitting or lying him with the legs higher than the head.
- Cover him but don't make him feel too warm
- Reassure the casualty by being calm and confident.
- For minor shock give casualty warm drinks
- Provide room for breathing and fresh air.
- Loosen tight clothing
- Dispose the casualty to the hospital.

7. **FAINTING (SYNCOPE)**

This is a complete loss of consciousness due to lack of oxygen in the brain.

Causes

- Injuries
- Severe pain
- Lack of food
- fatigue
- Horrifying sites
- Illness
- Bad or good news
- Lack of enough oxygen in the brain

Signs and symptoms

Dumpy and cold skin - Dry lips

Face becomes pale - Sweating

Shallow breathing - Yawning

- Gradual unconsciousness

Treatment

- Rest the casualty on his back with the feet slightly higher than the head to allow blood flow from the brain
- Loosen tight clothing
- Keep away the crowd
- Sit the casualty on the chair and put his head between the knees
- Hold his jaws and talk to him
- Fan the casualty to supply fresh air
- Place the casualty in a recovery position
- Dispose the casualty to the hospital

8. **ASPHYXIA (SUFFOCATION)**

This is a condition that a person suffer due to inadequate supply of oxygen in the blood (brain and tissue)

Causes

- Choking (obstruction of the airway)
- Strangulation
- Chest compression
- Electrical injury
- Poisoning
- Paralysis in cases of a stroke
- Air pollution
- Drowning
- Carbon monoxide.

Signs and symptoms

- Rate and depth of breathing is rapid and noisy
- Shallow breathing with froth
- Congestion around the neck
- Fits may occur
- Restlessness
- Unconsciousness

- Pale lips
- Increased heart rate
- Dizziness
- Casualty points at the mouth and the throat.

Treatment

- Remove the casualty from the cause
- Open the airway
- Loosen tight clothing
- Reassure the casualty
- Raise legs higher than the head
- Dispose the casualty to the hospital.

Methods of emergency resuscitation

1. Mouth to mouth resuscitation (Kiss of life)

Lie the casualty facing up

Tilt the head backwards

- Hold the jaws
- Breathe to the mouth several times until you see movement of the chest. The exercise to continue up to a period of 10 minutes.

Why mouth to mouth may not be possible

Sometimes

- Incase the casualty is producing body fluids through the mouth e.g. vomit blood.
- Incase there is serious injuries around in the mouth.
- Incase the casualty is suffering from contagious disease.
- Incase the casualty is facing downwards

2. Mouth to nose

3. Cardiac massage (Heart compression)

4. Nielsen Holger

ENVIROMENTAL INJURIES

High temperature

(a) Heat exhaustion

This condition occurs due to abnormal loss of salt and water in the body in a very humid environment. Water and salt is lost during vomiting and diahroearing.

Signs and symptoms

- Moist skin
- Fatigue
- Dizziness
- Pulse faster but weak
- Muscle cramp
- Pale skin.

Treatment

- Place the casualty in cool area.
- Loosen tight clothing
- Place feet slightly higher than the head.
- Treat for shock
- Dispose to the hospital.

(b) ***Heat stroke***

This is when the body is not able to loose excess heat.

Causes

Exposure to very high temperatures

Signs and symptoms

- Restlessness
- Unconsciousness
- Flushed skin, dry and burning
- Pulse is faster and strong
- Temperature extremely high.
- Fits/convulsions may occur.

Treatment

- Place in cool area
- Apply cold applications
- Fan the casualty
- Raise the legs slightly higher than the head
- Loosen tight clothing.

(c) Sun burn

This caused by severe burn of the sun

Signs and symptoms

Skin is red, hot, tender, swollen and blistering

Treatment

- Sponging
- Sips of cold water
- Place the casualty in a cold area.

Low temperature

(a) Frost bite

- Damage to the body tissue caused by exposure to extreme/severe cold. Blood vessels contract reducing the blood supply.
- Signs and symptoms
- Skin becomes extremely cold
- Skin becomes numb
- Skin change colour to greyish
- Skin feels hard and stiff
- Loss of function.

Treatment

- Remove all cold and wet cloths
- Place the body in warm place
- Remove tight items e.g. rings, belts e.t.c
- Give hot fluids
- Dispose the casualty to the hospitals

*(b) **Hypothermia***

This is dangerous lowering of the body temperature caused by prolonged immersion in cold water and general exposure to extreme cold.

Signs and symptoms

- Dizziness
- Severe headache
- Impair ness of muscular co-ordination
- Speech becomes blurred.

Treatment

Rap the casualty in a warm dry environment

Keep the victim a wake

Dispose the casualty to the hospital.

(c) Heat cramp

Caused by diminishing level of minerals in the body

Signs and symptoms

Painful muscles

- Nausea
- Diarrhoea
- Severe headache

Treatment

- Rest the casualty in a comfortable position
- Massage the muscles
- Place the casualty in a cool area
- Replace the fluid in the area.

<u>BONE INJURIES</u>

1. Fracture

This is cracking or breaking of the bone

- Causes
- Direct force
- Indirect force
- Muscular contraction/action.

Description of fractures

(a) Green stick

This is cracking of the bone due to body weight

Mostly common with children.

(b) ***Cross fracture***

The bone is cut across
(c) ***Oblique fracture***
The angle of breakage is 45^0
(d) ***Spiral fracture***
The line of breakage has spiral shape
(e) ***Quashed fracture***
The bone is severely damaged
(f) ***Avulsion***
The bone moves out of its position

Types of fractures
1**.** ***Simple/closed***
This is when the bone is injured and the surrounding tissue is not interfered with i.e. no would occurs.
2. ***Open/compound complicated***
This is where the broken bon penetrates through the skin injuring other body tissues e.g. muscles and blood vessels
3. ***Multiple fracture***
This is when body suffers more than one fracture.
Bone is broken at more than one point.

Signs and symptoms
Swelling and bruises
Pain
Tenderness
Deformity.

- Lose of movement and co-ordination
- Shock
- Bleeding
- Pale skin
- Temperature rises

Treatment
- Treat for shock
- Stop bleeding
- Treat for wounds
- Put the affected part on the splinters
- Immobilize the affected part
- Dispose the casualty to the hospital

JOINT INJURIES
1. Sprain

This is when the tendons and ligaments at the joints are torn or twisted due to sudden turning of the joint in a natural movement.

Signs and symptoms

- Tenderness
- Loss of movement
- Severe pain
- Change in skin colour
- Rise in temperature
- Treatment
- Support the joint in a comfortable position
- Treat for shock
- Elevate injured part
- Apply cold ice to control swelling, reduce pain e.t.c
- Apply firm bandage at the joint
- Rest the joint with no active movement
- Dispose the casualty to the hospital

2. Dislocation/luxation

Displacement of one or more bones at the joint i.e bones out of the alignment.

Causes

- Direct force
- Indirect force
- Muscular contraction/action

Signs and symptoms

- Bones not in the right place (deformity)
- Pain over the joint
- Bleeding due to tear of the muscles
- Swelling
- Affected part sensitive pressure and pain
- Tenderness
- Loss of movement
- Pale skin

SKIN/MUSCLES INJURIES

1. **Wound**

This is a break in continuity of the body tissue e.g. skin.

Classification of wounds

(a) ***Open wounds***: this is when there is a break in the skin.

(b) ***Closed wound***: this is when there is injury on the underlying tissue.

Types of open wounds

(i) ***Abrasion***

Caused by friction on the surface of the body. Skin may be removed discoloration of the area and bleeding e.g. sliding and tackling in soccer.

(ii) ***Incision/clear cut***

The line of the cut is straight. Produced by sharp cutting objects like knife, razor e.t.c
(iii) ***Laceration/torn***
This is irregular breaking or tearing of the tissue caused by a great force exerted on the body e.g. by barbed wire.
(iv) ***Puncture/ stabs***
Caused by sharp objects piercing the skin creating a hole in the tissue e.g. stepping on a thorn or nail, being stabbed with a knife.
(v) ***Avulsions***
This occurs when a tissue is forcibly separated or torn from the victim's body. There is a lot of bleeding.
(vi) ***Contusions/ bruises***
This capillary bleeding beneath the unbroken skin. This is caused by a knock or a blow by blunt instrument. The blood vessels are torn and the tissues are damaged.

Signs and symptoms

- Severe pain
- Swelling
- Rise in temperature
- Discoloration
- Skin becomes pale
- Treatment
- Rest in a comfortable position
- Cold application

(vii) *Burns and scalds*
Burns are caused by a dry fire or heat.
Scalds are caused by moist heat.

General treatment for all wounds
Remove all the impurities.

Wash the wound with warm soapy water
(Should be running water).
Cover the wound with dressing e.g. cotton or gauze
Place the bandage on the dressing
Immobilize the affected part.
Stop any bleeding
Treat for shock
Dispose the casualty to the hospital.

2. **Bleeding**
This is oozing of the blood from the wound or injury.
Types of bleeding
i. **External bleeding**
a) *Arterial bleeding*
The blood spurts out and is bright red colour due to plenty of oxygen.

b) *Venous bleeding*
The blood flows out smoothly and is dark red in colour due to less oxygen.
c) *Capillary bleeding*
The blood oozes out and is bright red in colour.

Pressure points
These are where arteries near the bone and can be pressed to stop bleeding e.g. wrist/radial; base of the neck/carotid, temple, elbow/radial vein, base of the thigh/groin.

Signs and symptoms of external bleeding
- Face/lips become pale
- Skin cold and dumpy
- Casualty feels fainty/dizziness
- The pulse is rapidly becoming weaker
- Restlessness
- Shallow breathing
- Yawning
- Thirsty

Treatment
- Direct pressure on the affected part
- Indirect pressure on the affected part
- Cold applications e.g. ice cubes, cold spray
- Elevation of the affected part to reduce the blood flow to the affected part.
- Cover the wound with sterilized dressings
- Immobilize the affected part
- Dispose the casualty to the hospital.

ii) *Internal bleeding*
This is oozing of blood from internal body organs/tissues without break of the skin.

Signs and symptoms
- Swelling on the affected part.
- Blood coming out from nose, ears and other openings.
- Pain
- Blood shot eyes
- Skin is dark in colour
- Urine/stool containing blood.
- How the body naturally stops bleeding
- Lowering of the body temperature
- Blood pressure decreases
- Heart rate decreases
- Vasco constriction of the blood vessels
- Reduced supply of the blood to the affected area.

Artificial treatment
- Elevation of the bleeding part to reduce the blood flow to the wound

- Direct pressure on the affected part.
- Indirect
- Cold applications e.g. ice cubes
- Cover the wound with sterilized dressings
- Immobilize the affected part.
- Dispose the casualty to the hospital.

3. **Muscle cramp/muscle pull**

Muscle cramp is sudden contraction of the muscle fibres due to insufficient ions (salt) in the body.

Muscle pull is sudden overstretching of the muscle fibres.

Causes

- Lack of enough salt intake
- Excessive sweating
- Improper training
- Overhauling if muscles i.e. warming and cooling effect.
- Fatigue
- External forces e.g. knock, blow
- Inability to relax tight muscles
- Loose of function
- Swelling

Treatment

- Extend the affected muscles, pull both muscle to allow contraction
- Encourage adequate fluids and salt intake
- Rest the muscles in a comfortable position
- Apply cold application
- Massage the affected area

4. **Muscle tear**

This can either be complete tear or partial tear

Causes

- Direct force on the muscle
- Indirect force (tension force)
- A cut by sharp object

Signs and symptoms

- Pain
- Cramp
- Swelling
- Depression
- Loss of function of the affected part
- Discoloration: due to internal bleeding

Treatment

- Apply ice to reduce pain
- Immobilize the affected part
- Secure the affected part with bandage
- Dispose the casualty to the hospital

5**. Muscle soreness**

This is pain along the muscles due to accumulation of lactic acid.

6. **Muscle strain**

This is overstretching of a muscle beyond its limit.

7. **Blisters**

This is caused by rough surface/wearing of very tight shoes

Treatment

- Cold application
- Never peel the blisters

FIRST AID

This is the first medical treatment given to the casualty before disposing to hospital. It is a skilled application of treatment given on the occurrence of injuries or sudden illness suing the locally available materials and facilities.

First aid terminologies

a) *Signs*

This is what a first aider can be able to observe or feel.

b) *Symptoms*

These are sensations/feelings experienced by the casualty

c) *Diagnosis*

This is medical method of identifying a given injury

d) *Prognosis*

This is the chance of survival

e) *Treatment*

These are the efforts and measures taken to cure an illness/injury

f) *Rehabilitation*

This is the effect aimed at restoring the person to his normal place/restoring the injury part to its normal position.

Objectives of first aid

- To sustain or save life
- To prevent the condition from worsening
- To promote recovery
- Relieve pain

Qualities of a good first aider

- Should be in a position to use common sense
- Should be observant i.e. can be able to detect the signs and symptoms.
- Should be gentle i.e. avoid rough handling of the casualty.

- Should be resourceful i.e. should use locally available materials
- Should be tactful i.e. should be calm to avoid alarming the casualty.
- Should be explicit i.e. clear in the instructions giving.
- Should be confident
- Should be sympathetic
- Should be perseverance i.e. continue with the treatment to the end.
- Discriminative i.e. should be able to give priorities.

The scope of first aide

1**.** ***Assessing the situation***

Find out:

- The cause of the injury
- The dangers around
- Safety of both casualty and first aider.

2. Diagnosis

This involves identifying the injury

Methods of medical diagnosis

a***) Inspection (observation***)

Check for:

Bleeding, swelling, change of skin color, vomiting and tenderness.

Auscultation (listening)

Listen to:

Heartbeat, breathing

(c) ***Palpation (touching)***

You tough to feel:

Swellings, broken bones, rise in temperature, tenderness e.t.c

d) ***Tappin***g

You tap the chest to find out whether there is internal bleeding/water in the chest.

(e) Medical history of the casualty

You can check the available document; listen to the story narrated by the conscious casualty or story given by the people around.

Contents of first aid box

- Safety pins
- Dressings
- Bandages
- Pain killers
- Glucose/dextrose
- Ice cubes
- Tweezers
- Scissors
- Methylated spirit
- Clinical thermometer
- Disinfectant/antiseptic
- Jelly
- First aid book
- Ointments

Dressings

This is the immediate protective cover placed over the wound e.g. cotton wool and gauze

Uses of dressings

- To protect the wound
- To prevent germs from entering the wound
- To absorb discharge e.g. pus/blood
- Protect further swelling
- Stop bleeding.

Bandages

These are thin woven materials used to hold the dressings in place /putting the splints in position.

Examples include: gauze, elastic, adhesive, binder of muslin, emergency bandages e.g. string, triangular bandages.

Uses of bandages

- Used to hold dressings in position
- To support splints
- To stop further bleeding
- To carry the casualty
- To support the body part
- To immobilize the affected part.

NOSE BLEEDING

This is the loss blood from the body through the nose due to internal injury.

Causes

- High blood pressure
- Internal injuries to the head
- Sharp objects entering the nose
- Change in altitude
- Sneezing (forceful blowing of nose)

We appreciate your time

- **End**